

4.2) Customer Analysis

The Customer Analysis is the largest and most detailed module of this project. It is divided into four sub-sections:

- [Customer Demographics](#)
- [Window Shoppers](#)
- [Actual Buyer Customers](#)
- [Repeated Customers](#)
- [Top 50 Customers](#)
- [Churn Customers](#)

each addressing a specific business question about who our customers are, how they behave, and how much value they bring to the business.

• 4.2.1) Customer Demographics

Business Questions Answer about the customers demographics are as follows:

- 1) [What is the age distribution of our customer base, and which age segment represents the largest share of walk-in visitors?](#)
- 2) [Which age group demonstrates the highest conversion rate from walk-in visitor to actual buyer, and what does this reveal about our sales effectiveness across generations?](#)
- 3) [Which age group contributes the most revenue to the business, and is there a revenue concentration risk in any particular segment?](#)
- 4) [How are New and Referral customers distributed across different age groups, and which age group produces the most referral-driven customers?](#)
- 5) [Which customer type New or Referral generates higher revenue, and what does this indicate about the value of our referral network?](#)
- 6) ["Which payment method Cash, UPI, or Card is preferred by each age group, and does the mode of payment reveal any generational shift in spending and payment behaviour among our customers?"](#)
- 7) [What is the distribution of vision types: Near, Distance, Bifocal, and Progressive; across our entire customer base?](#)
- 8) [How does vision type vary between New and Referral customers, and does the type of vision problem influence how a customer arrives at our store?](#)

Why Customer Demographics Analysis Matters?

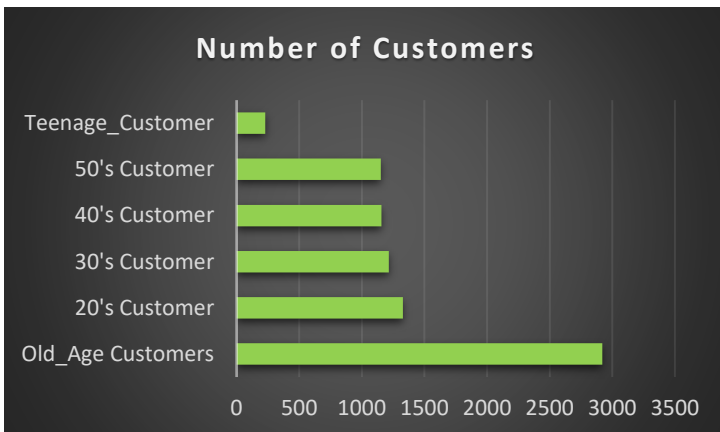
Understanding customer demographics is the foundation of every customer strategy. Before optimizing sales, marketing, retention, or referral programs, the business must first understand who its customers are, how they behave, and which customer segments contribute the greatest long-term value. This demographic analysis examines customer characteristics including age distribution, purchasing behaviour, referral patterns, payment preferences, and vision requirements to identify the primary customer segments driving business performance.

1) Count of Visited Customers across different Age groups

1) *What is the age distribution of our customer base, and which age segment represents the largest share of walk-in visitors?*

```
400
401 • SELECT
402 CASE
403     WHEN age < 20 THEN "Teenage_Customer"
404     WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
405     WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
406     WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
407     WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
408     WHEN age > 60 THEN "Old_Age Customers"
409 END AS Age_group,
410 COUNT(*) AS Count_of_Customers,
411 ROUND( COUNT(*) * 100 / 8000, 2) AS Percentage
412 FROM customers_details
413 GROUP BY Age_group WITH ROLLUP
414 ORDER BY Count_of_Customers DESC;
415
```

Age_group	Count_of_Customers	Percentage
Old_Age Customers	2920	36.50
20's Customer	1328	16.60
30's Customer	1214	15.18
40's Customer	1157	14.46
50's Customer	1152	14.40
Teenage_Customer	229	2.86



- Largest customer segment is Old Age Customers (above 60 years), accounting for 2,920 customers (36.5%)
- The second-largest segment is 20's Customers with 1,328 customers (16.6%), followed by 30's Customers (1,214; 15.2%), 40's Customers (1,157; 14.5%), and 50's Customers (1,152; 14.4%). These four age groups contribute relatively similar customer volumes.
- The smallest segment is Teenage Customers, with only 229 customers (2.9%)

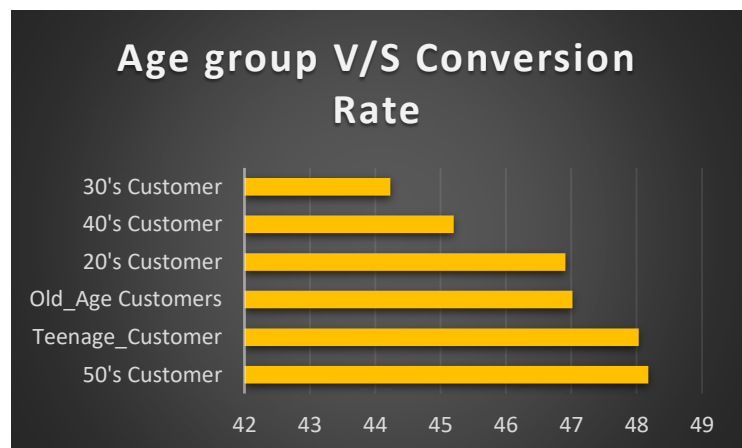
Business Insight:

- The data shows that the store's customer base is heavily concentrated in the 60+ age group, making senior citizens the most important target segment.
- Marketing campaigns, product inventory, and services such as progressive lenses, bifocals, and regular eye-checkup programs should continue to focus on this segment while parallelly exploring opportunities to attract more younger customers.

2) Customers Conversion Rate across different Age Groups

2) Which age group demonstrates the highest conversion rate from walk-in visitor to actual buyer, and what does this reveal about our sales effectiveness across generations?

```
SELECT
CASE
  WHEN age < 20 THEN "Teenage_Customer"
  WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
  WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
  WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
  WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
  WHEN age > 60 THEN "Old_Age Customers"
END AS Age_group,
ROUND( COUNT( DISTINCT i.customer_id)*100 /
COUNT( DISTINCT c.customer_id), 2) AS Conversion_Rate
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group
ORDER BY Conversion_Rate DESC;
```



The conversion rate across all age groups is remarkably consistent, ranging between 44% and 48% meaning roughly 1 in every 2 customers who walks into the store ends up making a purchase.

Result Grid		Filter Rows:
Age_group	Conversion_Rate	
50's Customer	48.18	
Teenage_Customer	48.03	
Old_Age Customers	47.02	
20's Customer	46.91	
40's Customer	45.20	
30's Customer	44.23	

Business Insights:

- The store is already performing well with older and mature age segments that is above 50 age.
- Although Teenage Customers represent the smallest visitor group (only 229 customers), they demonstrate one of the highest conversion rates (48.03%), indicating strong purchase intent once they enter the store.
- Also it's a strong signal that the sales staff is doing an excellent job at converting young walk-in visitors into actual buyers. (must take feedback from the staff, regarding the Genz customers behaviour and requirements in the store)
- There is a clear opportunity to increase footfall among younger and Gen Z customers. The high teenage conversion rate proves that once young customers walk in, the staff can close the sale.
- Hence, If our Future Goal is to enter into the Gen-z Retail Category here are the following solution and Risk mitigation:

Strategic Solution & Recommendation:

- To expand this segment, the business may need to invest in targeted marketing initiatives such as social media advertising, influencer collaborations, and digital engagement campaigns that resonate with younger consumers.
- The store may also need to maintain a more fashion-oriented inventory, including trendy frames, sunglasses, and eyewear collections that align with rapidly changing consumer preference

Risk Mitigation:

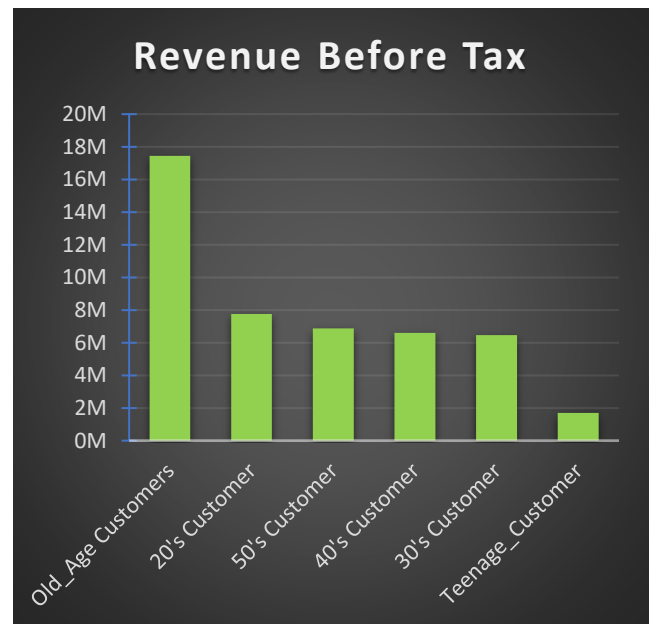
- However, entering this segment requires careful inventory management. Fashion trends among younger consumers evolve quickly, increasing the risk of slow-moving stock and inventory obsolescence.
- Therefore, the business should focus on faster inventory turnover, regular collection updates, and demand-driven purchasing decisions to minimize inventory risk while capturing growth opportunities in this market.

3) Contribution of different Age Groups in Revenue before Tax

3) Which age group contributes the most revenue to the business, and is there a revenue concentration risk in any particular segment?

```
SELECT
CASE
WHEN age < 20 THEN "Teenage_Customer"
WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN age > 60 THEN "Old_Age Customers"
END AS Age_group,
SUM(i.final_bill_amount) AS Revenue,
ROUND( SUM(i.final_bill_amount) * 100 / 46811413.00, 2)
AS Contribution_in_Revenue
FROM customers_details as c
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group WITH ROLLUP
ORDER BY Revenue DESC;
```

Age_group	Revenue	Contribution_in_Revenue
NULL	46811413.00	100.00
Old_Age Customers	17440454.00	37.26
20's Customer	7751143.00	16.56
50's Customer	6870009.00	14.68
40's Customer	6598649.00	14.10
30's Customer	6458010.00	13.80
Teenage_Customer	1693148.00	3.62



Old Age customers (age > 60) are the top revenue generators at approximately ₹1.8 Crore, which is a significant lead over every other age group.

The remaining five age groups 20s, 30s, 40s, 50s, and Teenage all fall within a relatively narrow band of ₹60 Lakhs to ₹80 Lakhs each, with 20s customers leading that mid-tier group at approximately ₹80 Lakhs.

Business Insights:

Positives: It validates that the business has built a strong and loyal base of older customers who trust the store, return regularly, and spend more per visit.

Negative: Over-dependence on a single age segment is a long-term business risk. As this customer base ages further, their purchasing frequency may decline

Meanwhile, the 20s, 30s, 40s, and 50s segments which together represent the majority of the customer count are each generating only ₹60–80 Lakhs individually.

The gap between Old Age revenue and everyone else is too wide to be comfortable from a business sustainability standpoint.

Strategic Solution & Recommendation:

The business needs to actively work on increasing the average transaction value and purchase frequency among the 20s, 30s, and 40s segments.

These are working-age customers with disposable income who are likely buying basic single vision lenses and standard frames. Introducing premium lens options, blue-light blocking glasses, and branded frame collections targeted at this demographic could meaningfully close the revenue gap between the Old Age segment and the rest.

Additionally, the Teen age segment, despite being the smallest in count, should not be overlooked.

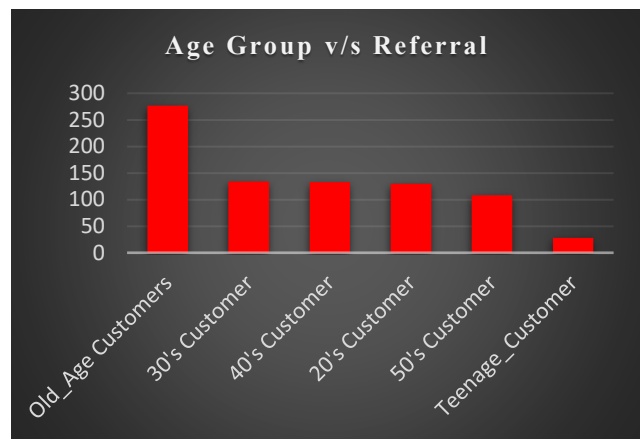
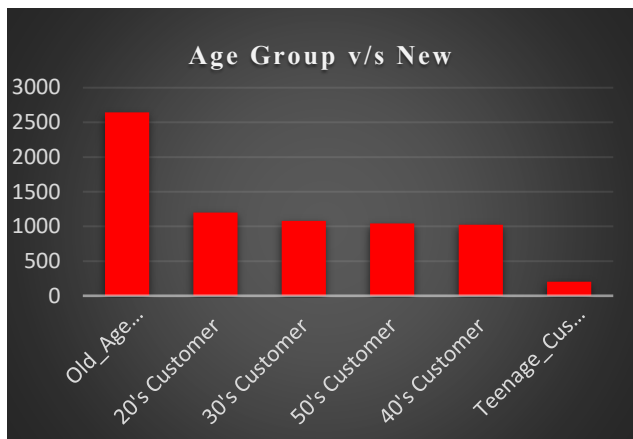
As discussed in the conversion rate analysis, teenage customers convert at a very high rate they simply do not walk in enough. Increasing their footfall alone could add meaningful revenue given how well the sales team handles this segment.

4) Customer Type across different Age Groups

4) How are New and Referral customers distributed across different age groups, and which age group produces the most referral-driven customers?

```
SELECT
CASE
WHEN age < 20 THEN "Teenage_Customer"
WHEN age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN age > 60 THEN "Old_Age Customers"
END AS Age_group,
SUM(CASE WHEN customer_type="New" THEN 1 ELSE 0 END)
AS Count_New,
SUM(CASE WHEN customer_type="Referral" THEN 1 ELSE 0 END)
AS Count_Refferal
FROM customers_details
GROUP BY Age_group;
```

Age_group	Count_New	Count_Refferal	Percentage_of_Referrals
20's Customer	1198	130	10.85
30's Customer	1080	134	12.41
40's Customer	1024	133	12.99
50's Customer	1044	108	10.34
Old_Age Customers	2643	277	10.48
Teenage_Customer	201	28	13.93
NULL	7190	810	11.27

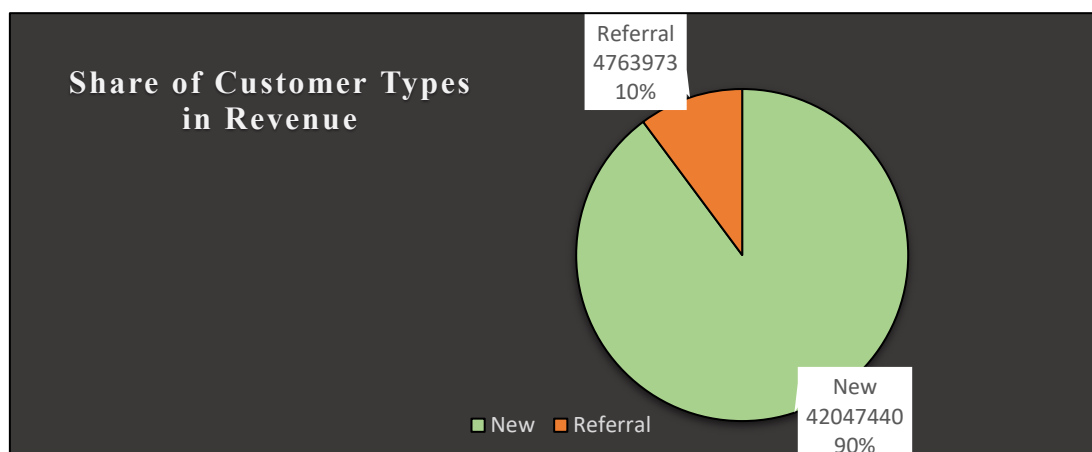


5) Revenue before Tax by different Customer Type

5) Which customer type New or Referral generates higher revenue, and what does this indicate about the value of our referral network?

```
SELECT c.customer_type,
SUM(i.final_bill_amount) AS Revenue,
ROUND(SUM(i.final_bill_amount)*100/
(SELECT SUM(final_bill_amount) FROM invoices), 2)
AS Revenue_Contribution
FROM customers_details as c
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY customer_type
ORDER BY Revenue DESC;
```

customer_type	Revenue	Revenue_Contribution
New	42047440.00	89.82
Referral	4763973.00	10.18



Business Insight:

- 1) The analysis shows that the business primarily acquires customers through New Customers, who account for approximately 90% of the total customer base, while Referral Customers contribute the remaining 10%.
- 2) From a volume perspective, the Old Age Customer (>60 years) segment records the highest number of referral customers, followed by customers in their 30s, 40s, 20s, and 50s, while teenage customers contribute the fewest referrals. However, this largely reflects the overall customer distribution, as the older age segment is also the store's largest customer group.
- 3) When referral activity is compared as a percentage within each age group, the differences are minimal. Most age segments have referral shares between 10% and 12%, indicating that referral behaviour is fairly consistent across generations rather than being concentrated within a specific demographic.
- 4) Overall, the findings suggest that referrals are generated uniformly across the customer base, with no single age group displaying significantly stronger referral behaviour. Although referral customers represent only a small share of total visitors, they remain an important customer acquisition channel, as demonstrated in the dedicated Referral Analysis.

Business Recommendation:

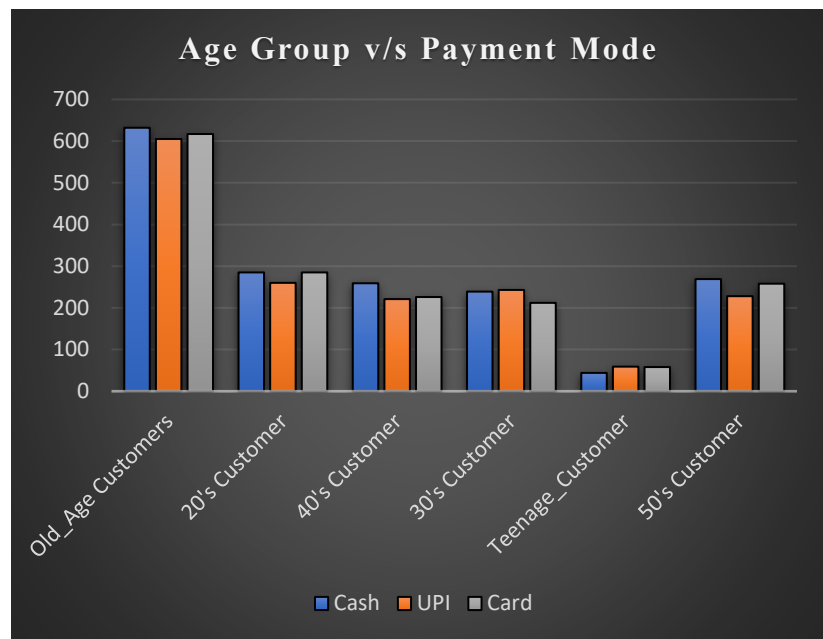
- Although Referral Customers represent only around 10% of the customer base, they should not be evaluated solely by their volume. As demonstrated in the dedicated Referral Analysis, this relatively small segment contributes approximately ₹47.64 lakh, representing 10.18% of the store's total revenue. This confirms that the referral network is a meaningful source of business growth achieved with virtually no direct marketing expenditure.
- Therefore, the business should continue investing in customer experience, trust-building, and referral initiatives to encourage satisfied customers to recommend the store. Even modest improvements in referral generation could produce a meaningful increase in customer acquisition while maintaining low acquisition costs.

6) Count of different Payments Mode usage by each Age group

6) Which payment method Cash, UPI, or Card is preferred by each age group, and does the mode of payment reveal any generational shift in spending and payment behaviour among our customers?

```
SELECT
CASE
WHEN c.age < 20 THEN "Teenage_Customer"
WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN c.age > 60 THEN "Old_Age Customers"
END AS Age_group,
COUNT( CASE WHEN i.payment_mode="Cash" THEN 1 END) AS Cash,
COUNT( CASE WHEN i.payment_mode="UPI" THEN 1 END) AS UPI,
COUNT( CASE WHEN i.payment_mode="Card" THEN 1 END) AS Card
FROM customers_details as c
INNER JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group;
```

Age_group	Cash	UPI	Card	Total
Old_Age Customers	632	605	617	1854
20's Customer	285	260	285	830
40's Customer	259	221	226	706
30's Customer	239	243	212	694
Teenage_Customer	44	59	58	161
50's Customer	269	228	258	755



payment_mode	Count_Payment_Modes
Cash	1535
UPI	1465
Card	1502

payment_mode	Revenue_Collected
Cash	16196941.00
Card	15674327.00
UPI	14940145.00

payment_mode	Count_Payment_Modes	Revenue_Collected	Contribution_in_Revenue	Average_order_value
Cash	1535	16196941.00	34.60	9373.229745
UPI	1465	14940145.00	31.92	9245.139233
Card	1502	15674327.00	33.48	9465.173309

Business Insight:

Understanding customers' preferred payment methods helps the business ensure a smooth checkout experience, optimize payment infrastructure, and reduce operational dependency on any single payment channel.

The analysis shows that the business has a well-diversified payment ecosystem. Cash, UPI, and Card transactions each contribute almost equally to the total number of payments and revenue collected.

Similarly, the Average Order Value (AOV) remains highly consistent across all payment methods (approximately ₹9,200–₹9,500), indicating that customers spend nearly the same amount regardless of how they choose to pay. This suggests that payment preference does not influence purchase value.

The age-group analysis also reveals a similar pattern. Across every customer segment, payment usage is relatively balanced, with no age group displaying a strong dependence on Cash, UPI, or Card. Even among senior customers (60+), transactions are distributed almost evenly across all three payment methods.

Business Recommendation

The business is not operationally dependent on any single payment method, which reduces payment collection risk and provides flexibility for customers during checkout. Going forward, the store should continue supporting all major payment options while ensuring a seamless payment experience across each channel rather than prioritizing one method over another.

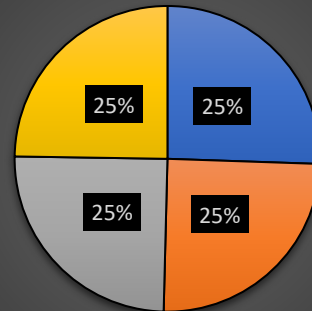
7) Distribution of Vision Types across all the Visited Customers

7) What is the distribution of vision types: Near, Distance, Bifocal, and Progressive; across our entire customer base?

```
-- Vision Type COUNT
SELECT p.vision_type,
COUNT(customer_id)
AS COUNT_Vision_Type,
ROUND( COUNT(customer_id)*100/
(SELECT COUNT(customer_id) FROM prescriptions), 2)
AS Percentage_of_Vision_Types
FROM prescriptions as p
GROUP BY p.vision_type
ORDER BY COUNT_Vision_Type DESC;
```

vision_type	COUNT_Vision_Type	Percentage_of_Vision_Types
Bifocal	2043	25.54
Near	1989	24.86
Distance	1988	24.85
Progressive	1980	24.75

Distribution of Vision Types



■ Bifocal ■ Near ■ Distance ■ Progressive

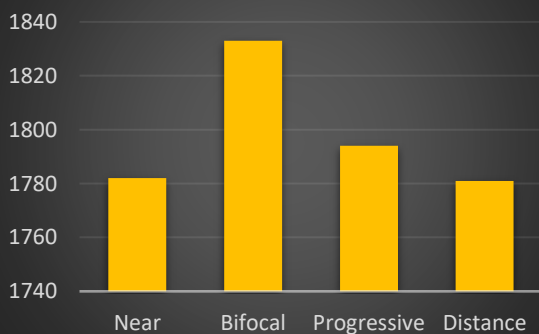
8) Count of Customer Type in different Vision type

8) How does vision type vary between New and Referral customers, and does the type of vision problem influence how a customer arrives at our store?

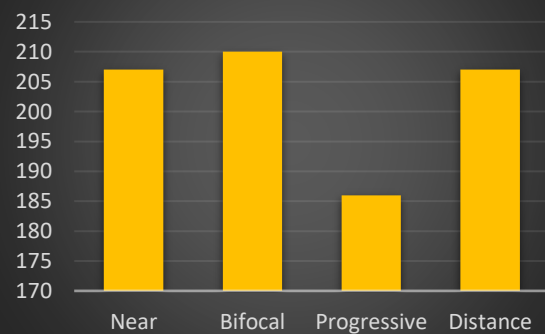
```
SELECT p.vision_type,
SUM(CASE WHEN c.customer_type="New" THEN 1 ELSE 0 END)
AS COUNT_New,
SUM(CASE WHEN c.customer_type="Referral" THEN 1 ELSE 0 END)
AS COUNT_Referral
FROM prescriptions as p
LEFT JOIN customers_details as c
ON c.customer_id=p.customer_id
GROUP BY p.vision_type;
```

vision_type	COUNT_New	COUNT_Referral
Near	1782	207
Bifocal	1833	210
Progressive	1794	186
Distance	1781	207

Vision Type v/s New



Vision Type v/s Referral



Business Insight

Understanding the distribution of customers' vision types is critical for an optical business because it directly influences clinical infrastructure requirements, inventory planning, staff expertise, and customer experience. Delivering an accurate prescription is the foundation of customer satisfaction, as even a minor prescription error (such as ± 0.25 di-opters) can result in blurred vision, eye strain, headaches, and ultimately loss of customer trust.

No single vision category dominates the customer base, indicating that the business must maintain expertise, inventory, and testing capabilities across all major vision correction requirements rather than specializing in only one category.

However, this finding should be interpreted alongside the previous demographic analyses. Earlier analysis identified customers above 60 years of age as the store's largest customer segment. This is particularly important because senior customers are more likely to require complex prescriptions such as progressive and bifocal lenses, and their eye prescriptions naturally change every two to three years due to age-related vision conditions such as presbyopia.

Although all vision categories contribute almost equally to the business, the large senior customer base makes prescription accuracy a strategic priority. Retaining these customers depends not only on product quality but also on consistently delivering precise vision correction during every visit.

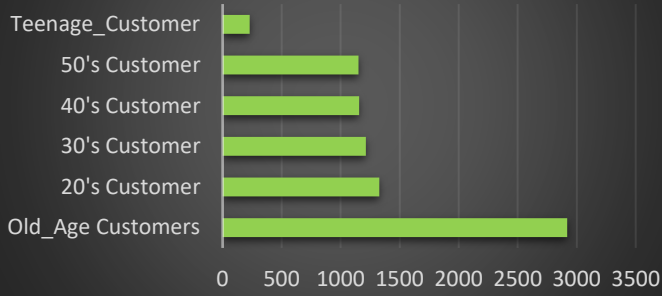
Business Recommendation

To support long-term customer retention and maintain a high-quality customer experience, the business should continue investing in its vision testing infrastructure. Key priorities include:

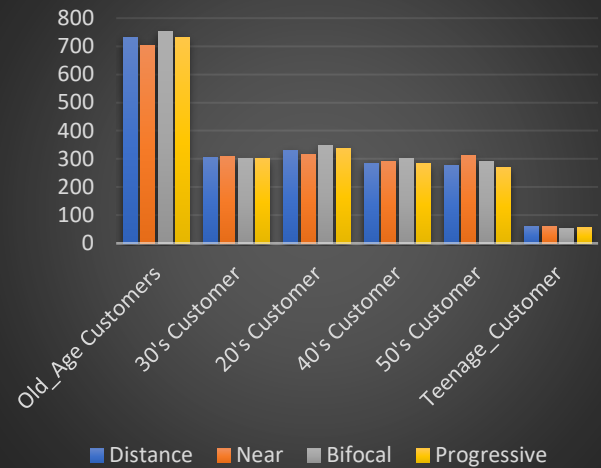
- Upgrading to high-precision digital eye examination equipment.
- Maintaining accurate PD (Pupillary Distance) measurement tools, particularly for progressive lenses.
- Hiring experienced optometrists capable of handling complex prescriptions.
- Ensuring skilled spectacle fitting to maximize comfort and visual accuracy.
- Conducting periodic calibration of testing equipment to minimize prescription errors.

1. Dashboard of Customer Demographics

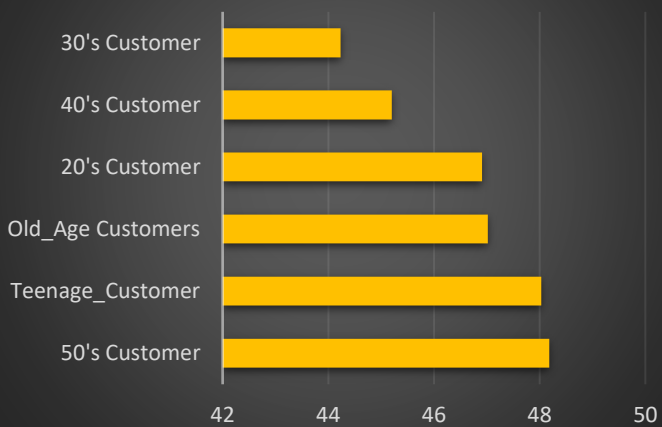
Number of Customers



Vision Type vs Age group



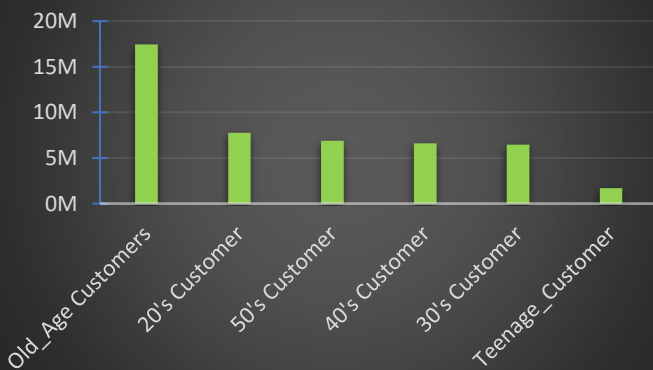
Age group V/S Conversion Rate



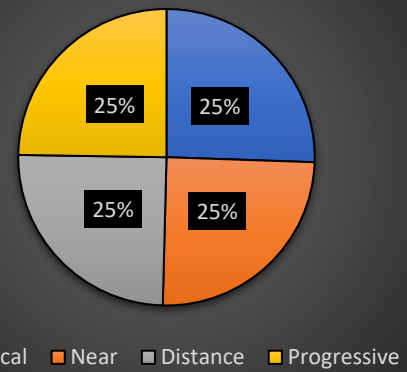
Share of Customer Types in Revenue



Revenue Before Tax



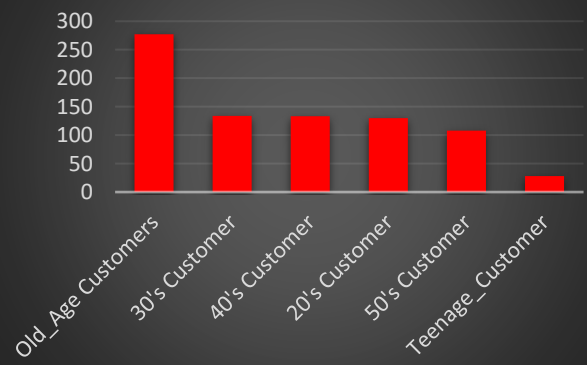
Distirbution of Vision Types



Age Group v/s New



Age Group v/s Referral



Customer Demographics Conclusion:

The Customer Demographics Analysis provides a comprehensive understanding of who the business serves before evaluating how customers behave throughout their purchasing journey. By analysing customer age, vision requirements, payment preferences, customer acquisition channels, and revenue contribution, we established the demographic profile of the store's customer base and identified the key segments driving business performance.

The analysis reveals that the business has developed a particularly strong presence among customers above 60 years of age, who consistently lead across customer volume, revenue contribution, and referral activity. Customers between 20 and 50 years also represent a stable and well-balanced customer base, while teenage customers, although smaller in volume, demonstrate encouraging conversion and retention characteristics that present future growth opportunities.

From an operational perspective, the business benefits from a well-diversified customer portfolio. Revenue is generated across multiple vision correction categories, payment methods are evenly distributed between Cash, UPI, and Card, and referral customers contribute a meaningful share of overall revenue despite representing only a small proportion of the customer base. These findings indicate that the business is not overly dependent on a single customer characteristic or payment channel, reducing operational risk while supporting sustainable growth.

One of the most significant observations throughout this section is the strategic importance of customer experience. Given the large proportion of senior customers and the nature of vision correction services, long-term success depends on delivering highly accurate prescriptions, maintaining customer trust, and providing exceptional service quality. As vision prescriptions naturally evolve over time particularly

1. Customer Demographics



Who are our customers?

2. Customer Analysis



How do they purchase?

3. Churn Analysis



Who are we losing?

4. Staff Analysis



Who is driving business performance?

5. Referrer Analysis



How are we acquiring customers organically?

6. Product Analysis



What are customers buying?

7. Final Business Conclusion



What should the business do next?

- Overall, the Customer Demographics Analysis establishes the foundation for the remainder of this report. While it explains who the customers are...
- The following sections will examine how they behave including purchasing patterns, customer conversion, repeat purchases, churn behaviour, referral networks, and customer lifetime value to identify actionable opportunities for improving business performance.